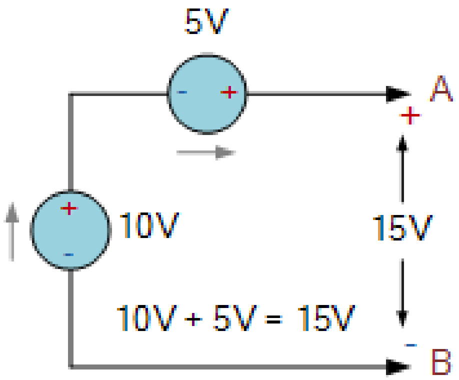
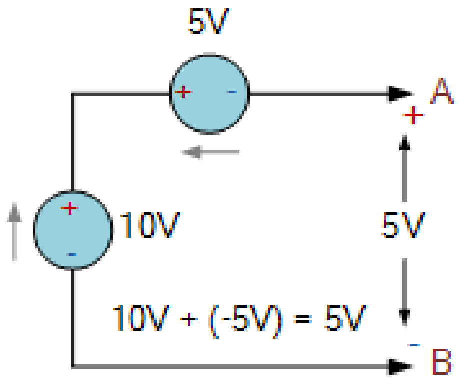
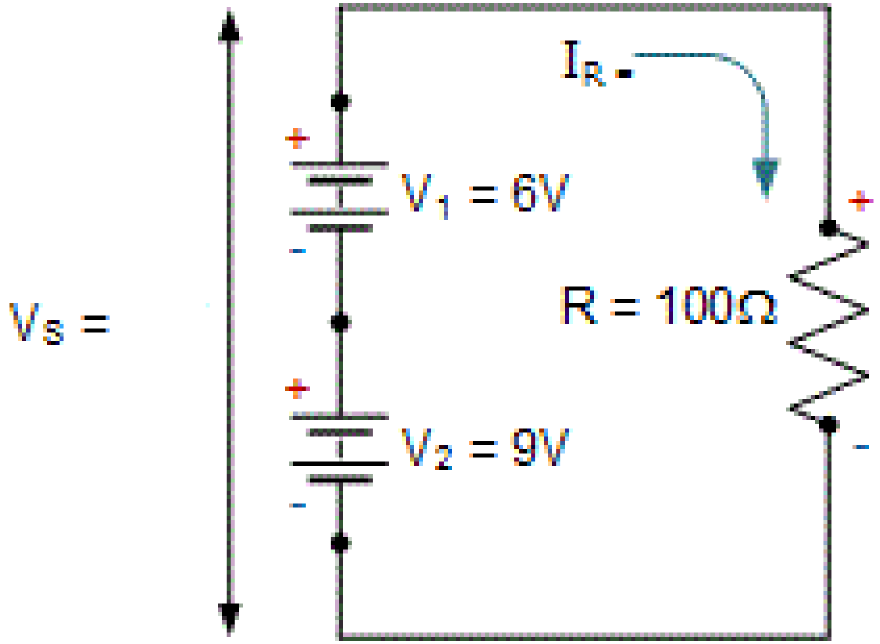
Voltage and Current Sources contd.



Series Aiding Voltages
(Voltage Addition)

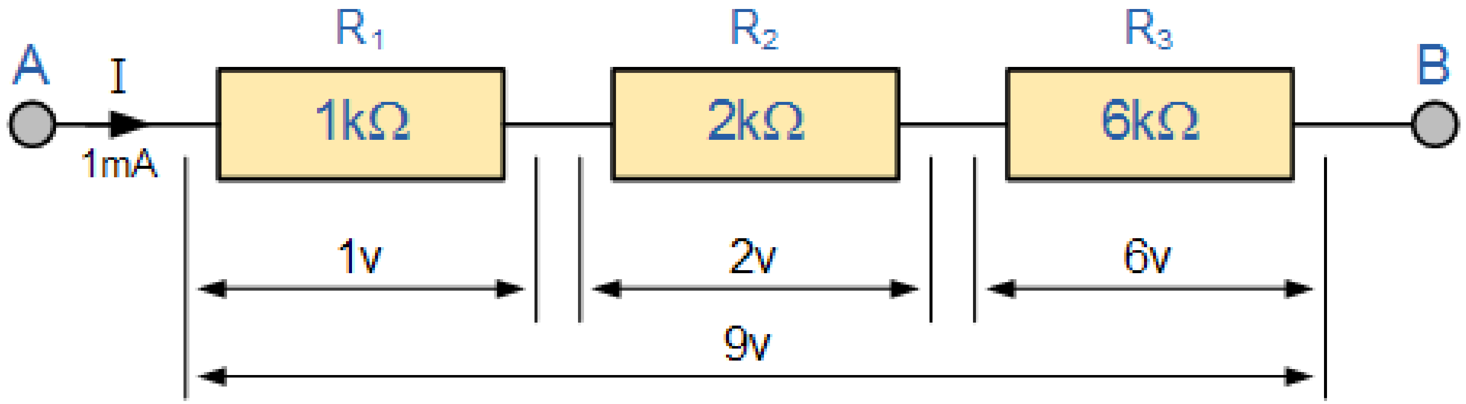


Series Opposing Voltages
(Voltage Subtraction)



Series and Parallel Resistor Comb.

- Resistors are said to be connected in "Series", when they are daisy chained together in a single line
- Resistors in series has common current flowing through them



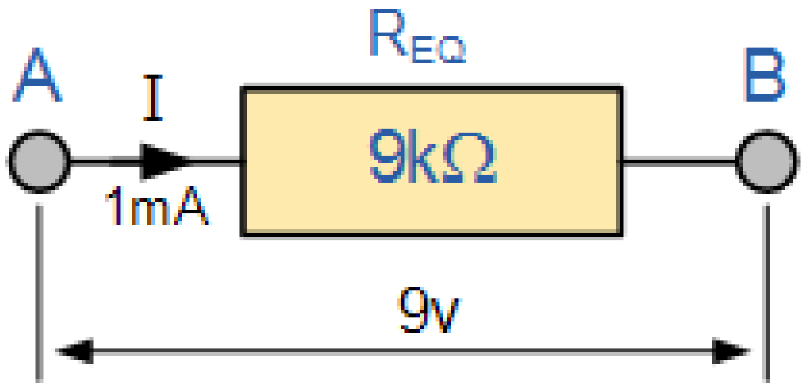
Series and Parallel Resistor Comb. Contd.

- The amount of current will remain same throughout the network

$$I_T = I_{R1} = I_{R2} = I_{R3}$$

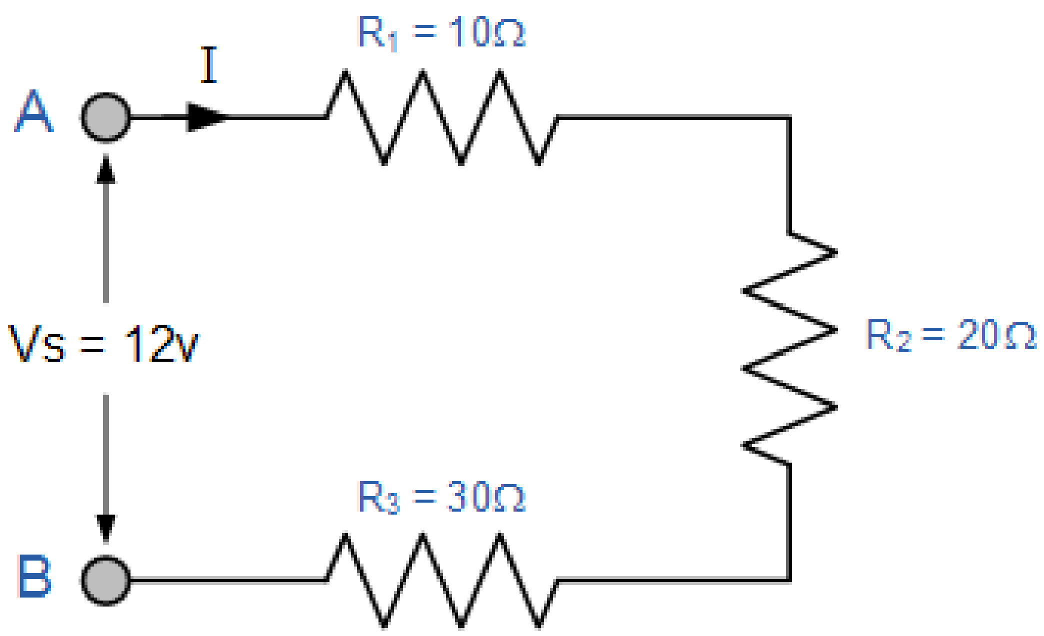
- The equivalent resistance is the sum of all the resistance

$$R_T = R_1 + R_2 + R_3$$



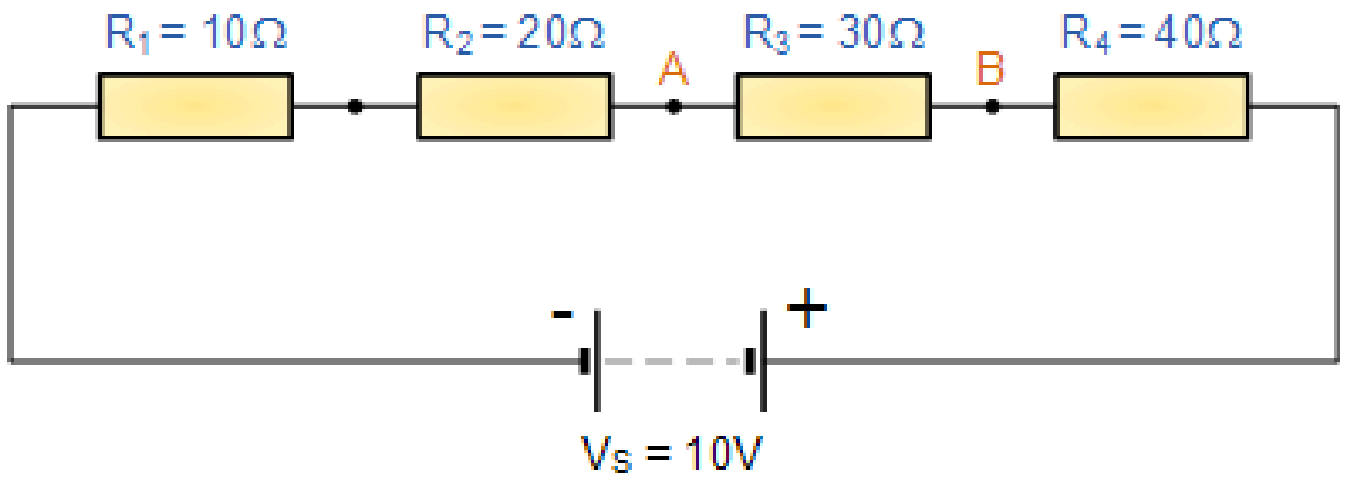
Series and Parallel Resistor Comb. Contd.

- A simple example for calculating the total resistance and current



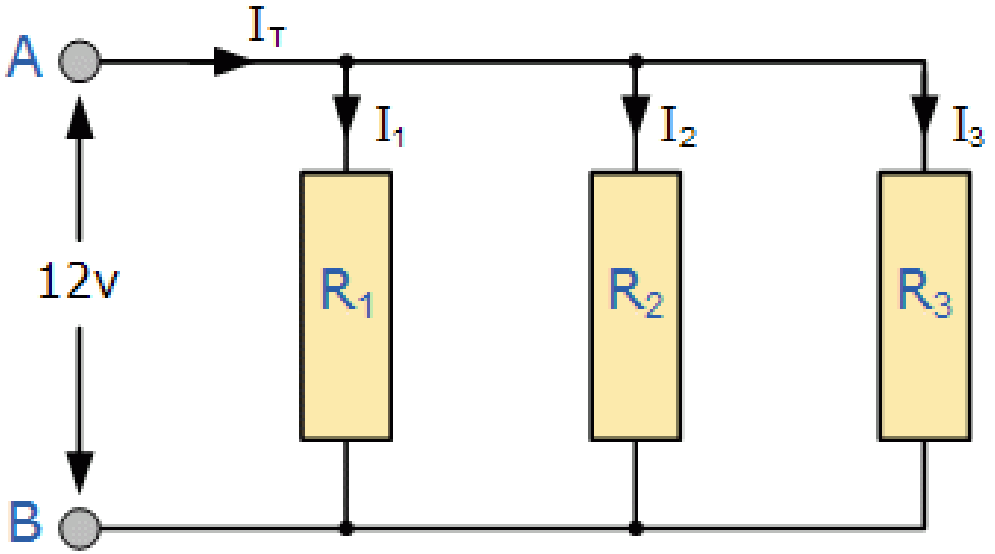
Series and Parallel Resistor Comb. Contd.

- Another example for finding the voltage between two points



Series and Parallel Resistor Comb. Contd.

- In a parallel resistor network the circuit current can take more than one path as there are multiple paths for the current
- Resistors in Parallel have a Common Voltage across them but current will divide (depends upon the resistance value)



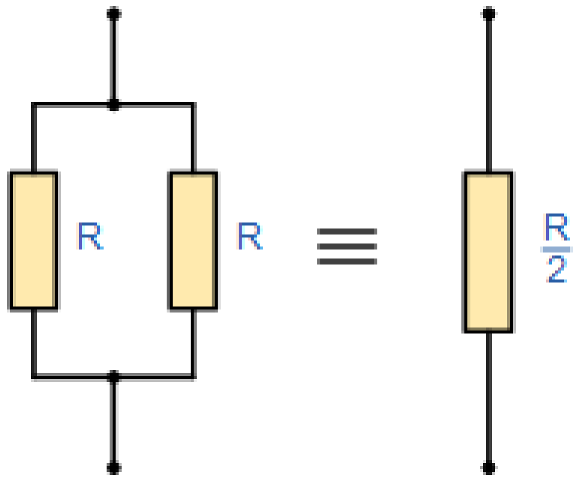
Series and Parallel Resistor Comb. Contd.

- The amount of voltage will remain same throughout the network

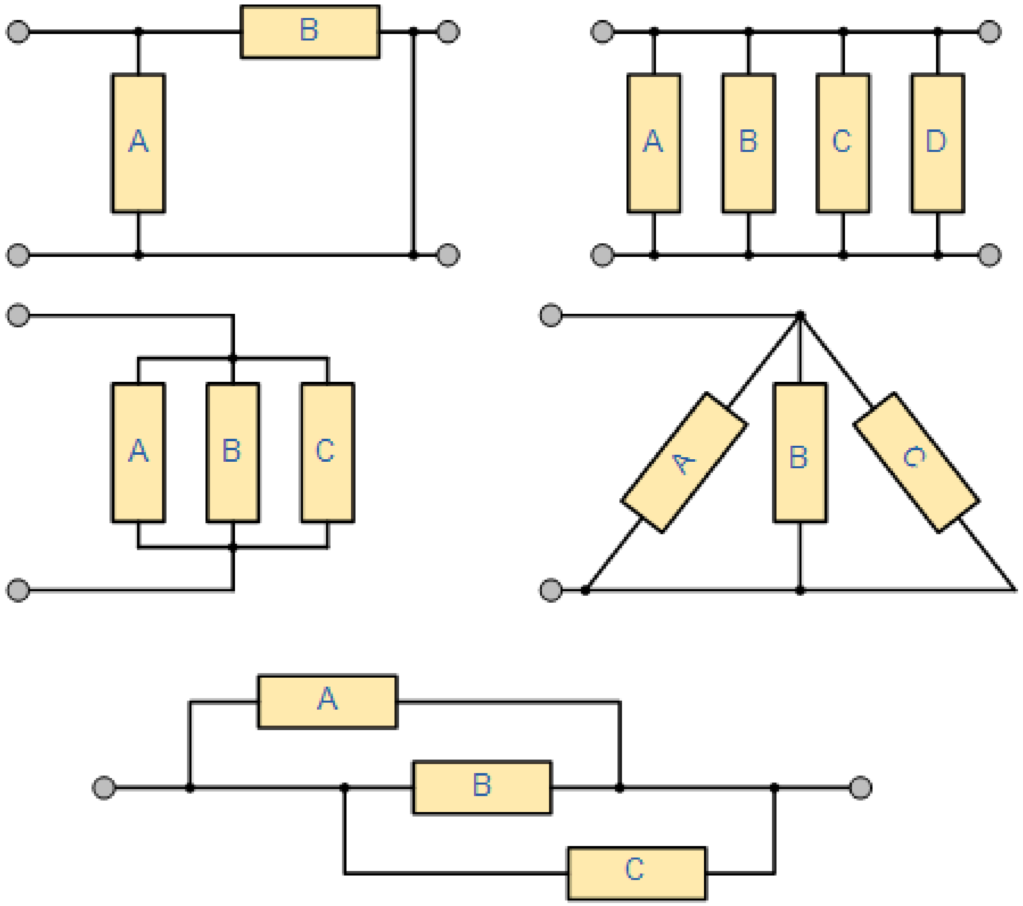
$$V_T = V_{R1} = V_{R2} = V_{R3}$$

- The equivalent resistance can be calculated as follows

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

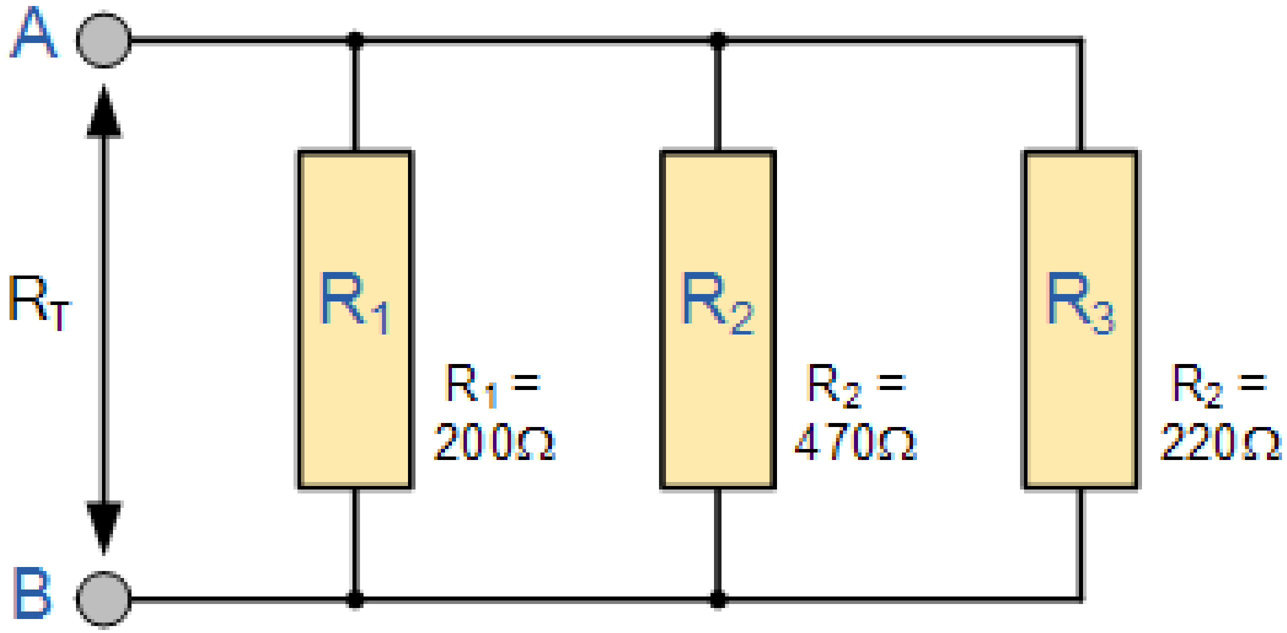


Series and Parallel Resistor Comb. Contd.



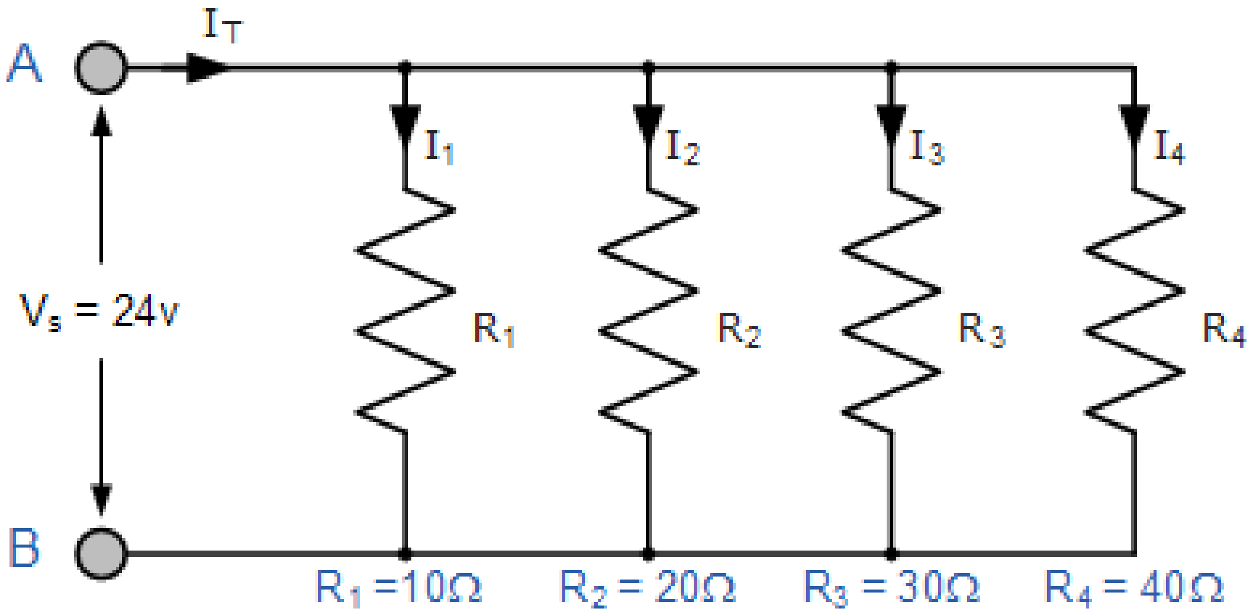
Series and Parallel Resistor Comb. Contd.

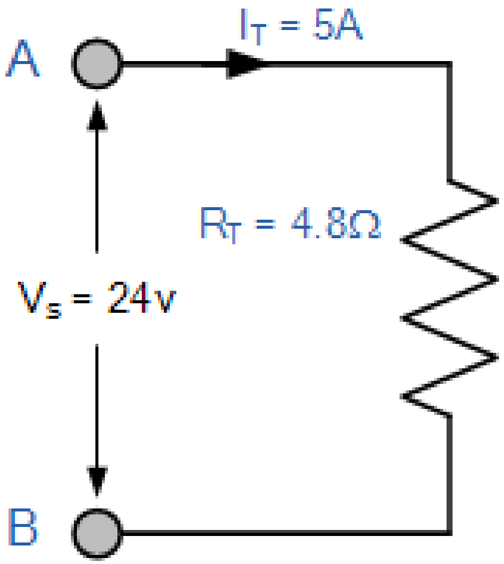
- An example for parallel network



Series and Parallel Resistor Comb. Contd.

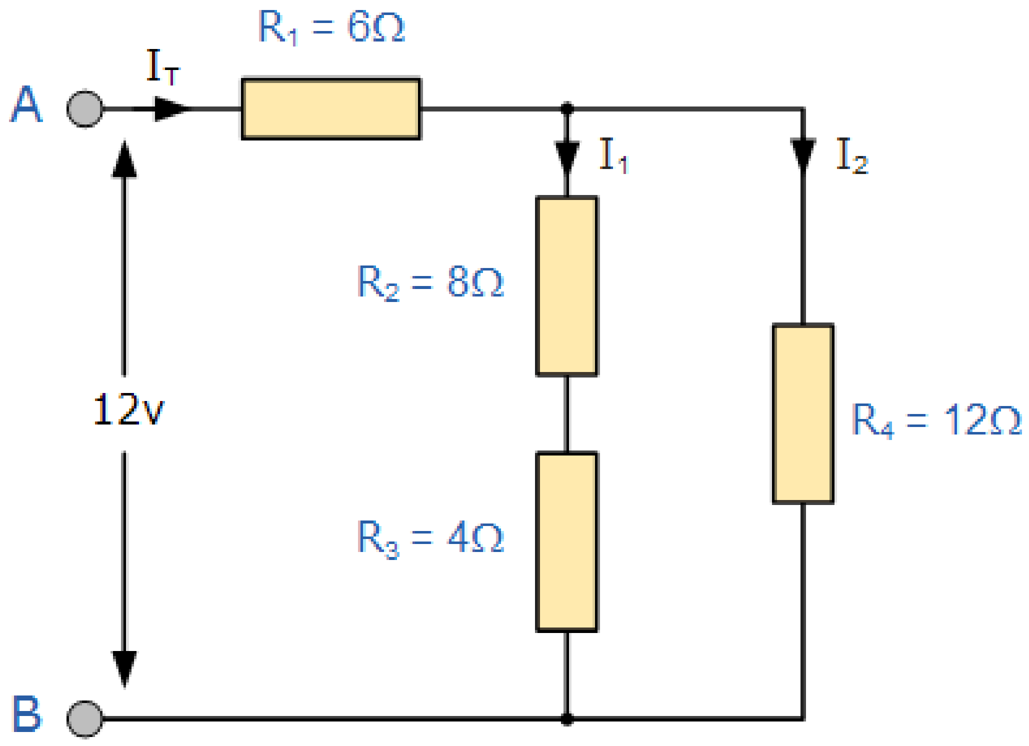
- Another example





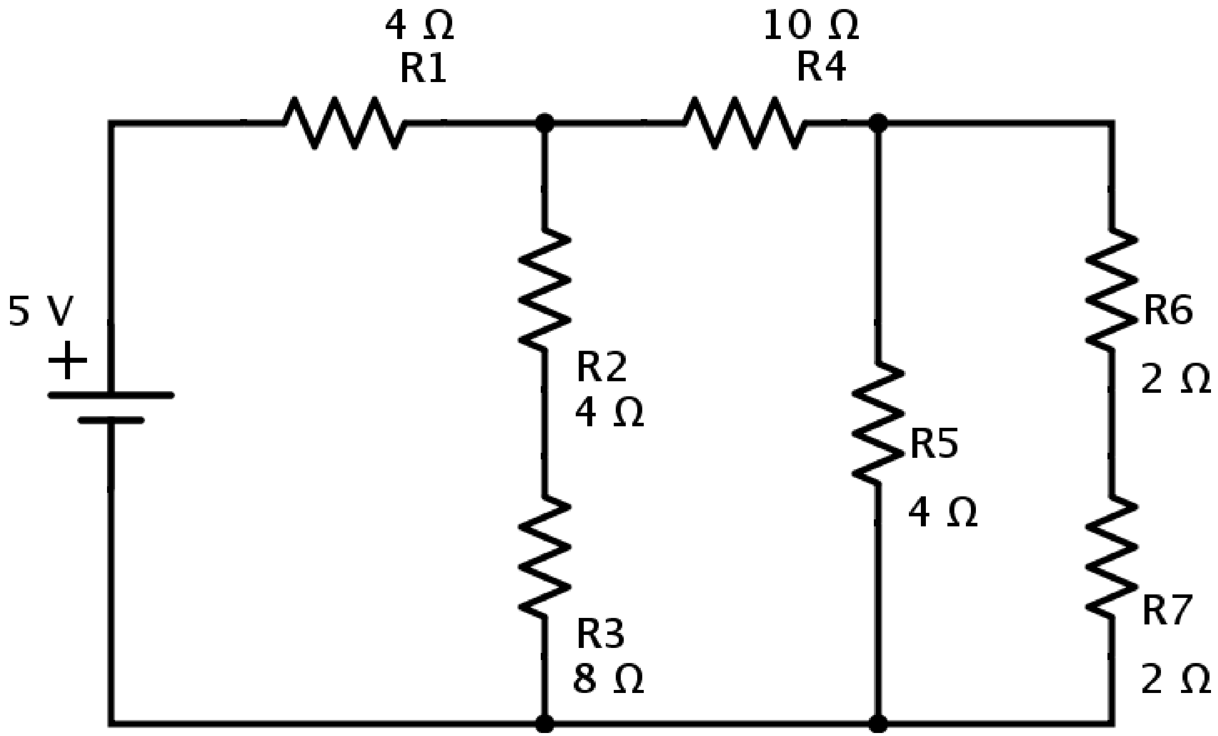
Series and Parallel Resistor Comb. Contd.

- A little complex example



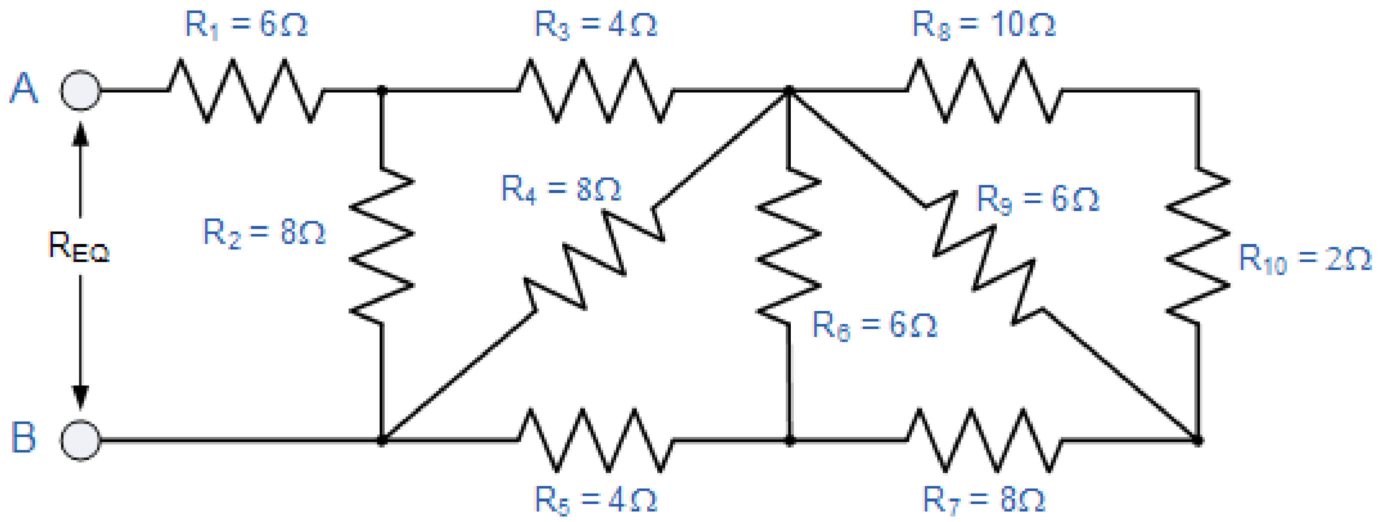
Series and Parallel Resistor Comb. Contd.

- Task for you



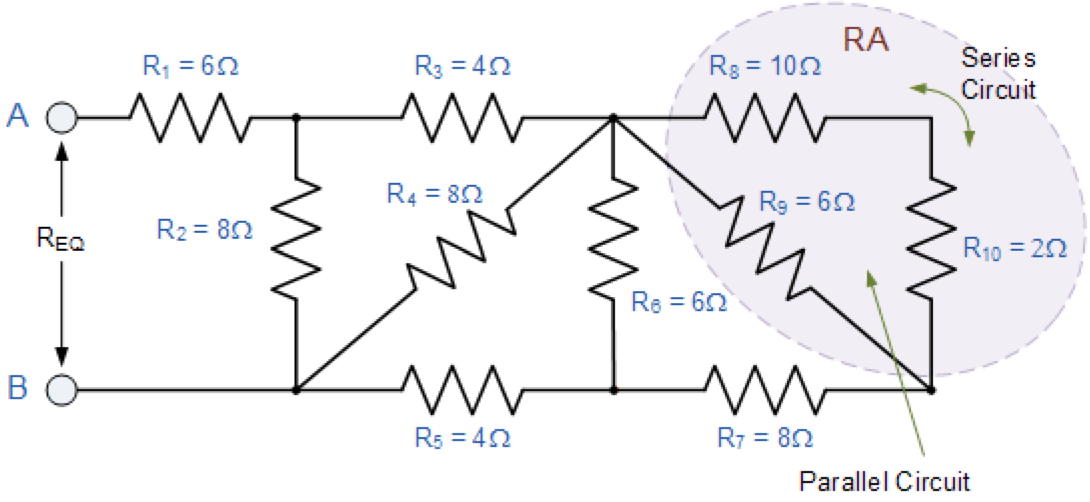
Series and Parallel Resistor Comb. Contd.

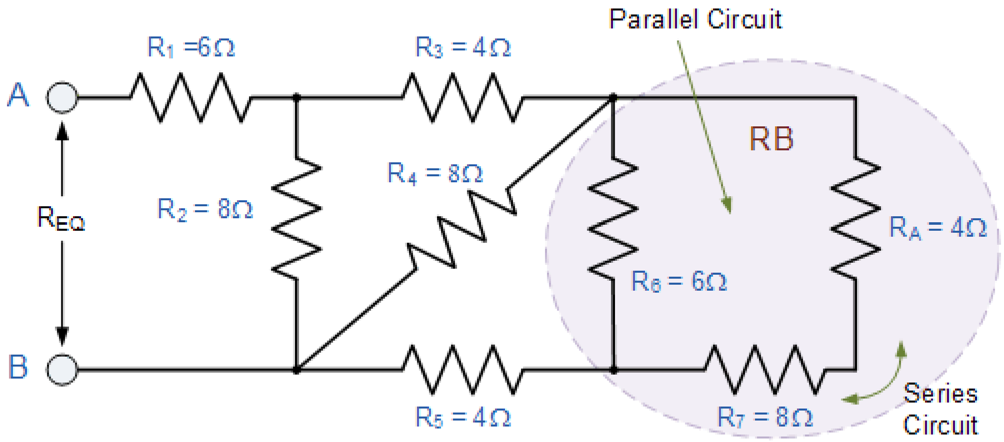
- A complex example



Series and Parallel Resistor Comb. Contd.

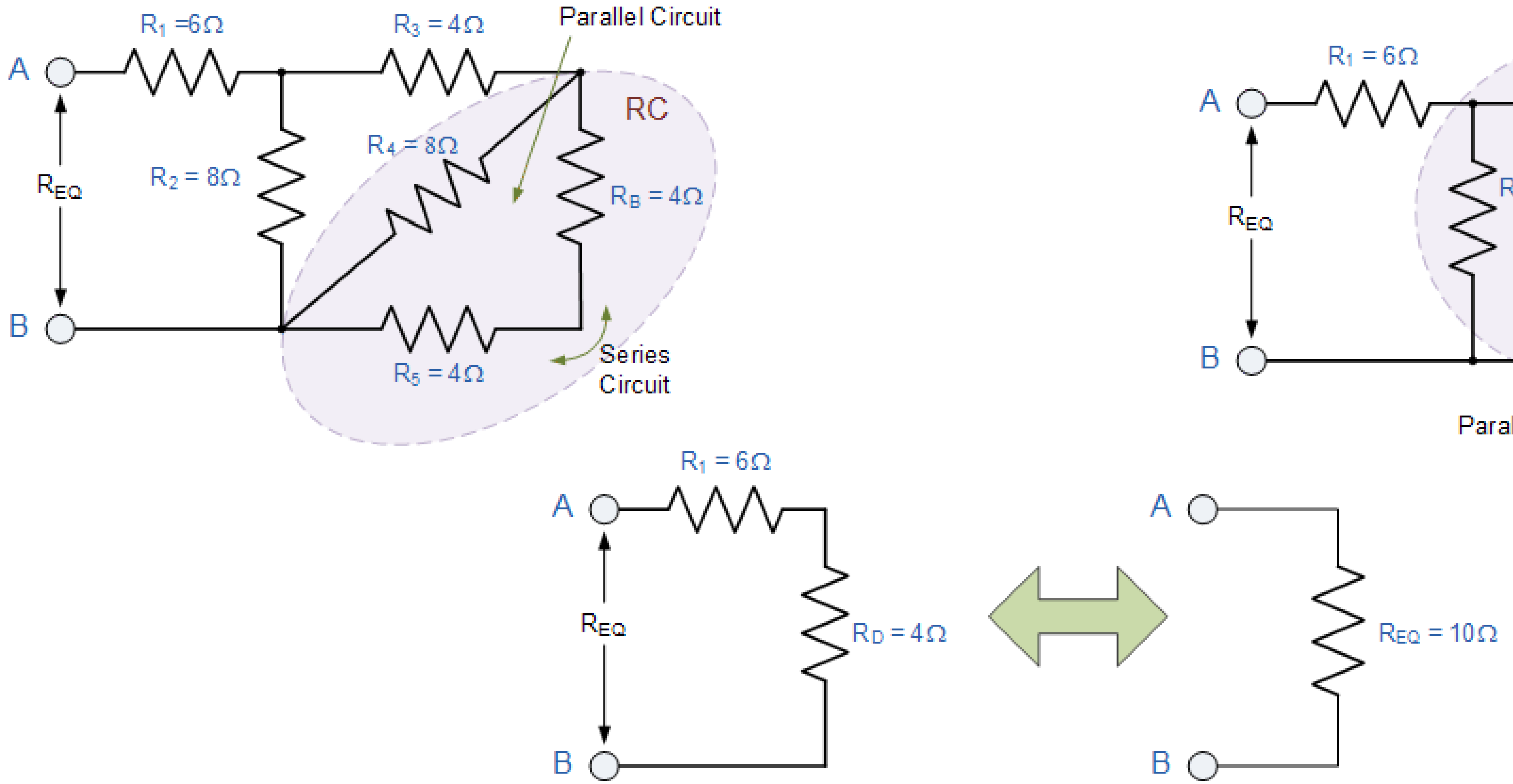
- A complex example





Series and Parallel Resistor Comb. Contd.

- A complex example



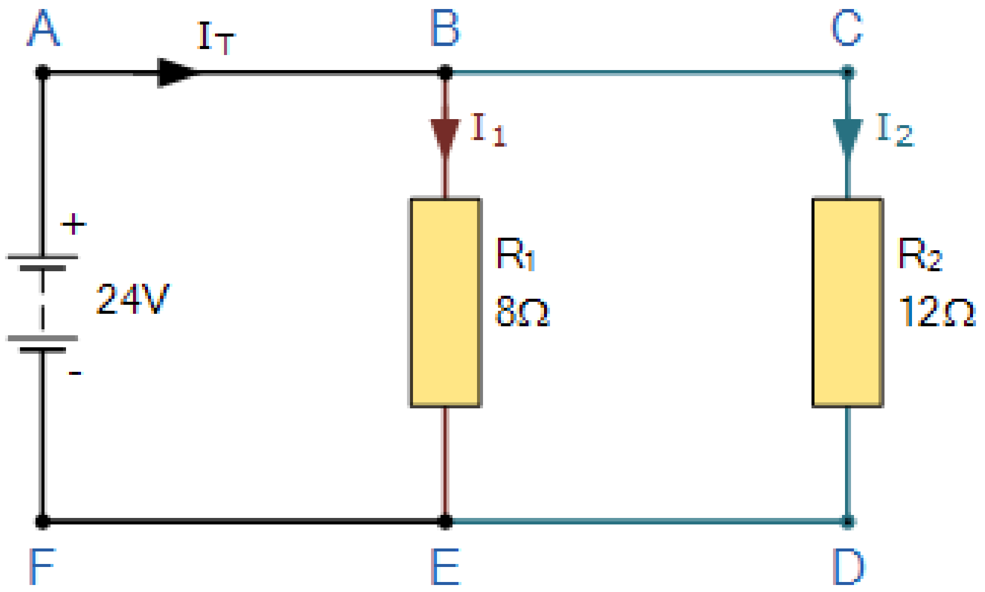
Kirchhoff's Current Law (KCL)

- Kirchhoff's Current Law is one of the fundamental law used for circuit analysis
- It states that the total current entering a circuits node is exactly equal to the total current leaving the same node
- Mathematically we can write it as

$$\sum I_{IN} = \sum I_{OUT}$$

Kirchhoff's Current Law (KCL) contd.

- lets take a simple example



Kirchhoff's Current Law (KCL) contd.

- lets take a complex example